

Program: Diploma in Fire Technology and Safety	
Course Code: 3463	Course Title: Occupational Health & Safety
Semester: 3	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:3 T:1 P:0)	Periods per semester: 60

Course Objectives:

- To introduce the concepts of occupational health and the hazards associated with the workplace.
- To familiarize students with industrial toxicology and the prevention of occupational diseases.
- To provide knowledge on industrial hygiene, ergonomics, and the control of physical and chemical hazards.
- To equip students with an understanding of personal protective equipment and basic epidemiology in occupational health

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Principles of Safety Management		Principles of Safety Management	2

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Identify occupational health hazards and the importance of wellness.	15	Understanding
CO2	Analyze the role of industrial hygiene in preventing health risks.	14	Understanding
CO3	Identify chemical and biological hazards and control measures in the workplace.	15	Understanding
CO4	Recognize the importance of personal protective equipment (PPE) and epidemiology in health management.	14	Understanding
	Series Test	2	

CO–PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2	3	2			
CO2	3	3	2	2			
CO3	3	3	2	3			
CO4	3	3	3	2			

3-Strongly mapped,2-Moderately mapped,1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Understand the concepts of occupational health, health hazards, and methods of disease prevention.		
M1.01	Introduction to Occupational Health and Wellness.	7	Understanding
M1.02	Notifiable occupational diseases, effects, and prevention.	8	Understanding
Contents: Introduction to Occupational Health and Occupational Health Hazards Mind-body communication for wellness. Occupational health hazards and prevention of notifiable diseases. Community health and role of occupational health services. Levels of disease prevention and degenerative illnesses.			
CO2	Understand industrial hygiene, physical hazards, and ergonomic issues in the workplace.		
M2.01	Role of Industrial Hygienist in hazard recognition and evaluation.	7	Understanding
M2.02	Hazard control methods: noise, vibration, and radiation.	7	Understanding
	Series Test–I	1	
Contents: Industrial Hygiene and Physical Hazards Introduction to industrial hygiene and the role of industrial hygienist. Work-related heart diseases, dermatoses, hearing loss, and eye disorders. Industrial noise hazards and control methods. Vibration hazards and effects on the human body. Types of radiation and preventive measures. Ergonomics: assessing and controlling MSDs and cardiovascular fatigue. Thermal stress and heat disorders.			

CO3	Learn about chemical hazards, air sampling, and biological hazards in the workplace.		
M3.01	Chemical hazards: gases, vapors, solvents, and particulate matter.	7	Understanding
M3.02	Biological hazards: microorganisms and bio aerosols.	8	Understanding
Contents: Chemical and Biological Hazards Mechanisms of breathing and gaseous exchange. Recognizing and evaluating chemical hazards. Biological hazards: microorganisms and bioaerosols. Control measures for chemical and biological hazards.			
CO4	Understand the importance of PPE and epidemiology in preventing occupational health issues.		
M4.01	Types and selection of personal protective equipment (PPE).	7	Applying
M4.02	Fundamentals of epidemiology and exposure assessment strategies.	7	Applying
	Series Test–II	1	
Contents: Personal Protective Equipment (PPE) and Epidemiology Non-respiratory and respiratory PPEs: types and selection. Basic principles of epidemiology in occupational health. Exposure assessment strategies and psycho-social risk factors. Biological rhythms and shift work considerations.			

Text/Reference

T/R	Book Title/Author
T1	Encyclopaedia of Occupational Health and Safety - Jeanne Mager Stellman (ed), International Labour Office (1998).
T2	The Industrial Environment - DHHS (NIOSH) Publication (1973).
R1	Patty's Industrial Hygiene and Toxicology - Clayton, C.D., and Clayton, F. (1981).
R2	Environmental, Safety, and Health Engineering - Gayle Woodside & Dianna Kocurek (1997).
R3	Accident First Aid - Cantlie, James (1932).